**26900147** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)

## REGULAR EXAMINATION October 2026

## Fifth SEMESTER

## Discipline Specific Elective (DSE) - MG5DSEBCA304 - DIGITAL IMAGE PROCESSING

(2024 ADMISSION ONWARDS)

**Duration: 2 hours** **Maximum Marks: 70**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Explain image sampling and quatization. [U] / [CO1]
2. Differentiate between an analog image and a digital image. [An] / [CO1]
3. Analyze the effect of dilation on an object. [An] / [CO3]
4. Identify one application of hit-or-miss transformation. [U] / [CO3]
5. State any two characteristics of Rayleigh noise. [K] / [CO4]

**[2x5 = 10]**

### Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

6. Explain different color models used in image processing. [U] / [CO1]
7. Analyze different fundamental steps in Digital Image Processing [U] / [CO1]

8. Write short note on [U] / [CO1]
   a) Image processing hardware
   b) Image processing software
   c) Mass storage

9. State the objectives of histogram equalization. Explain how it improves the contrast of a low-contrast image. [U] / [CO2]

10. Explain the 2-D DFT and its inverse transform. Describe image representation in the frequency domain. [An] / [CO2]

11. Analyze how opening and closing operations can be used for noise removal. [An] / [CO3]

12. Analyze how Otsu's method determines the optimal thresholding . [An] / [CO4]

**[6x5 = 30]**

## Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

13. For two pixels P(x,y)and Q(s,t) calculate their Euclidean, D4 and D8 distances for given coordinate values. [A] / [CO2]

14. Analyze the fundamentals of morphological image processing. Discuss its basic concepts, terminology, and major applications with suitable examples. [An] / [CO3]

15. Compare region growing and region splitting and merging techniques. [An] / [CO4]

**[15x2 = 30]**